Repeatability-Calibrated EMG–Kinematic Discordance in Pediatric Gait: A Cross-Sectional and Longitudinal Pilot Study with Wearable Sensors

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***Abstract*—Clinical gait assessment in children with cerebral palsy (CP) and idiopathic toe walking (ITW) often treats joint kinematics and surface electromyography (EMG) as separate outcomes, and rarely asks whether a deviation in one exceeds the ordinary trial-to-trial variability of the measurement. We propose an interpretable, phase-specific concordance analysis that scores ankle dorsiflexion/plantarflexion and area-normalised tibialis anterior and gastrocnemius envelopes against a typically developing (TD) reference, calibrates each score with a within-session two-trial repeatability threshold and a leave-one-out TD background rate, and assigns every visit-limb to one of four classes: both normal, ankle-only, EMG-only or both abnormal. Applied to the public wearable-sensor dataset LIVE-GaitNeuroKids (19 CP, 11 ITW and 25 TD children, 179 trials), ankle-only abnormality dominated in CP (53.2% of visit-limbs versus 10.2% in TD; difference +43.0 points, child-level permutation $p = 0.001$), and the result held under one-vote-per-child weighting, age matching and removal of sensor offset (43.5% versus 6.1%, $p = 0.003$). EMG-only abnormality was not elevated in CP, and an ITW excess driven by swing-phase gastrocnemius activity (19.0% versus 8.2%) did not reach significance. When abnormality was additionally required to exceed same-day variability, no EMG-only case remained in any group. Treatment-cycle-aware pairing yielded only seven valid pre/post children, of whom one showed robust kinematic improvement without EMG change. Area-normalised EMG shape deviations are therefore mostly smaller than within-session noise, which limits their use for tracking treatment response.**

***Index Terms*—cerebral palsy, idiopathic toe walking, surface electromyography, gait analysis, inertial sensors, measurement repeatability**

I. Introduction

Cerebral palsy (CP) and idiopathic toe walking (ITW) are common causes of equinus and toe-walking gait in children referred for clinical gait analysis. Treatments such as botulinum toxin injection, nerve ablation and casting aim to change both how the ankle moves and how the calf and shin muscles are activated. Kinematic outcomes are usually summarised with deviation indices such as the Gait Deviation Index and the Gait Profile Score [1], [2], and EMG has analogous indices that measure the distance of activation envelopes from a typical reference [3]. These measures are, however, usually reported side by side rather than jointly, so the clinically interesting case in which the two modalities disagree is not made explicit.

Disagreement between movement and muscle activation is plausible. In a randomised trial, botulinum toxin with rehabilitation improved gait kinematics in children with CP without improving surface EMG patterns [4], and in a cohort of 51 patients the relation between kinematic abnormality and abnormal activation patterns was not direct [5]. Detecting such disagreement in an individual child is difficult because pediatric EMG is variable: within-session waveform variability in children is roughly twice that of adults [6], and repeatability baselines are needed before a between-visit change can be called real [7]. A discordance flag that ignores this variability risks labelling noise as a clinical finding.

The recently released LIVE-GaitNeuroKids dataset [8], [9] provides synchronised inertial kinematics and four-channel surface EMG from children with CP, ITW and TD controls, two walking trials per session and repeated clinical assessments around treatment. This makes it possible to calibrate a concordance analysis against same-day repeatability using wearable rather than laboratory instrumentation. The contributions of this paper are:

- a visit-level, phase-specific four-class concordance analysis of ankle kinematics and tibialis anterior/gastrocnemius EMG, calibrated by a within-session two-trial repeatability threshold and a leave-one-out TD background rate;
- to our knowledge, the first such analysis on LIVE-GaitNeuroKids, with child-level inference, age matching and a sensor-offset sensitivity analysis;
- evidence that area-normalised EMG shape deviations rarely exceed same-day variability, together with a treatment-cycle-aware pairing of the dataset's longitudinal visits that exposes how few valid pre/post pairs it contains.

II. Related Work

Gait deviation indices. The Gait Deviation Index and Gait Profile Score summarise kinematic distance from a TD ref-

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erence [1], [2]. EMG analogues include the Kerpape-Rennes EMG-based gait index and the EMG Profile Score, the latter being a mean per-muscle root mean square error (RMSE) from TD envelopes, and synergy-based dynamic motor control indices [3], [10]. Our per-muscle RMSE is closely related to the EMG Profile Score; the novelty here lies in the joint classification and its calibration, not in the score itself.

EMG and kinematics after treatment. Kinematic improvement without EMG pattern change after botulinum toxin [4], an indirect relation between EMG and kinematic abnormality [5], and systematic post-surgical EMG change despite stable motor-control indices [3] all motivate analysing the two modalities jointly.

Repeatability. Within-session EMG variability in healthy children exceeds adult values [6], and inter-visit EMG repeatability in CP is comparable to that of unimpaired individuals [7]. Standard error of measurement and minimal detectable change are the usual tools for separating real change from measurement error [11]. We use the simplest within-session analogue that two trials allow.

Toe walking. Premature swing-phase gastrocnemius activity has been reported in both mild diplegic CP and ITW [12], and botulinum toxin can normalise ankle EMG timing in ITW [13].

III. Materials and Methods

A. Dataset

LIVE-GaitNeuroKids (Zenodo v1.1, CC BY 4.0) [8], [9] contains 59 children (22 CP, 12 ITW, 25 TD). Kinematics were recorded with a lower-body inertial system at 100 Hz and surface EMG from bilateral tibialis anterior (TA) and medial gastrocnemius (GM) placed following SENIAM [14]. The providers band-pass filtered (20–400 Hz), rectified and low-pass filtered (6 Hz) the EMG, synchronised it with the kinematics, segmented gait cycles and resampled each to 101 samples, with per-sample support (foot contact) labels. Each session contains two walking trials over a 12 m path. CP and ITW children were assessed immediately before treatment (T0), 1–2 weeks after (T1) and about three months after (T2). We used only these processed cycles. In total, 179 trial files from 19 CP, 11 ITW and 25 TD children loaded; 19 limb records whose kinematic sheet lacked that limb's ankle angle were excluded, leaving 339 trial-limb records.

B. Signal preparation

For each trial and limb we used the sagittal ankle angle and the TA and GM envelopes. Each EMG cycle $e(t)$ was divided by its area, $\tilde{e}(t) = 101\, e(t) / \sum_t e(t)$, so that 1.0 equals the cycle-mean activation and only the shape (timing) of activation is compared. Cycles were averaged within a trial. Because the kinematic and EMG sheets could contain different numbers of cycles, the stance/swing boundary was set per trial-limb to the median first sample at which the support label switched from contact to no contact (median 52% of the cycle, range 35–81%). For hemiparetic and monoparetic CP only the affected limb was analysed; both limbs were analysed for diparetic CP, ITW and TD.

TABLE I
Within-session repeatability thresholds δ_s and TD normal limits U_s

	δ_s (95th pct.)		U_s (TD 95th pct.)	
Score	Stance	Swing	Stance	Swing
Ankle (deg)	5.47	5.22	9.44	12.08
TA (norm.)	0.72	0.67	0.59	0.85
GM (norm.)	0.47	0.70	1.03	0.70

C. TD reference and deviation scores

A mean waveform was computed for each TD child over all trials and limbs, and the reference was the unweighted mean across children, so each child contributes equally (Fig. 1). For channel $c \in \{\text{ankle}, \text{TA}, \text{GM}\}$ and phase $\phi \in \{\text{stance}, \text{swing}\}$ the deviation score is

$$d_{c,\phi} = \sqrt{\frac{1}{|\phi|} \sum_{t \in \phi} \big(x_c(t) - r_c(t)\big)^2}, \tag{1}$$

giving six scores per trial-limb (ankle in degrees, EMG in units of cycle-mean activation). TD children were scored against a leave-one-out reference that excludes them. Visit-limb scores are the mean over the visit's trials.

D. Repeatability threshold and normal limit

For every visit-limb with at least two trials (151 visit-limbs: 45 TD, 75 CP, 31 ITW), we took the absolute score difference between the first two trials. The within-session repeatability threshold δ_s for each score s is the 95th percentile of these differences, pooled over groups (Table I). The normal limit U_s is the 95th percentile of TD visit-limb scores; for a TD child it is recomputed without that child.

E. Four-class concordance

A visit-limb is ankle-abnormal if either ankle score exceeds U_s and EMG-abnormal if any of the four EMG scores exceeds U_s. Crossing the two flags gives four classes: both normal, ankle-only, EMG-only and both abnormal. Under the stricter *robust* definition a score must exceed $U_s + \delta_s$, that is, lie beyond the normal limit by more than same-day variability. Applying the same rule to TD children with leave-one-out limits gives the background rate against which patient groups are compared; because six scores are tested, this rate is expected to be well above 5%.

F. Statistical analysis

Class proportions are reported over visit-limbs with 95% confidence intervals from a cluster bootstrap over children (2000 resamples) [15]. Differences from TD are tested with a child-level permutation test (5000 permutations of group labels between children). Three checks follow: (i) a one-vote-per-child estimate, the mean over children of each child's class share; (ii) within TD, Spearman correlation of age with each score; (iii) an age-matched re-analysis restricted to patient visits within the TD age range (5.0–12.2 years). In a sensitivity analysis each ankle curve and the reference were mean-centred

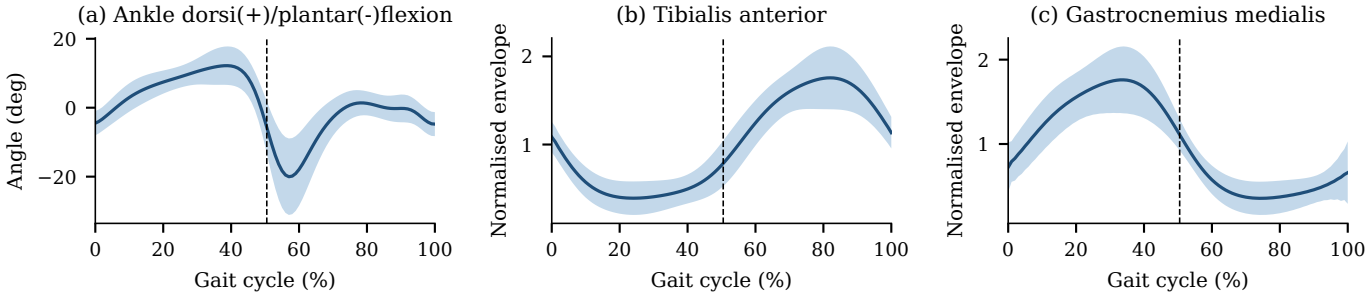


Fig. 1. TD reference waveforms from 25 children (line: mean of per-child means; band: ±1 SD between children; dashed line: median toe-off). EMG is area-normalised so that 1.0 is the cycle-mean activation.

before scoring, so a constant sensor offset cannot create an ankle deviation; all thresholds, limits and classes were recomputed.

G. Longitudinal pilot

Many children were treated more than once and the T0/T1/T2 labels restart with each treatment, so in most children the latest T2 preceded the latest T0. Using the treatment dates in the participant records, a T0 was paired with the follow-up (T2, else T1) of the same treatment cycle only when T0 fell between the previous and current treatment and the follow-up between the current and next treatment. A limb was *discordant* if at least one ankle score decreased by more than δ_s while no EMG score decreased by more than δ_s.

IV. Results

A. Reference waveforms

The TD reference reproduced the expected pattern (Fig. 1): GM activity peaked in stance (33% of the cycle) and was low in swing, TA was active around initial contact and peaked in swing (82%), and peak plantarflexion occurred just after toe-off (57%).

B. Cross-sectional concordance

Table II and Fig. 2 give class proportions. TD visit-limbs fell outside the normal limits 20.4% of the time, which is the background rate. In CP, ankle-only abnormality was the dominant class (53.2%), whereas EMG-only abnormality (8.1%) matched the TD rate. In ITW the ankle was rarely abnormal, and EMG-only abnormality was 19.0%. Among ITW EMG-abnormal visit-limbs, 64% were flagged by the GM swing score and 55% by the TA stance score; in CP, 71% were flagged by the TA stance score.

Under the robust definition every EMG-only case in CP and ITW disappeared, and only CP ankle-only abnormality remained substantial (41.9%). For most EMG scores the threshold δ_s was of the same order as, or larger than, the margin by which patient scores exceeded U_s (Table I); the median TA stance visit score, for example, was 0.40 in CP and 0.32 in TD (all limbs) against $\delta_s = 0.72$.

TABLE II
Concordance classes (% of visit-limbs, 95% child-bootstrap CI). n: visit-limbs/children; Std.: score exceeds U_s; Rob.: exceeds $U_s + \delta_s$

Def.	Class	CP (n=62/19)	ITW (n=42/11)	TD (n=49/25)
Std.	Both normal	24.2 [9.0, 43.9]	71.4 [58.8, 84.1]	79.6 [66.7, 91.7]
	Ankle-only	**53.2** [30.8, 71.3]	2.4 [0.0, 7.5]	10.2 [2.0, 20.4]
	EMG-only	8.1 [1.6, 17.3]	**19.0** [8.6, 30.0]	8.2 [2.0, 16.0]
	Both abn.	14.5 [5.1, 26.8]	7.1 [0.0, 20.0]	2.0 [0.0, 6.2]
Rob.	Ankle-only	41.9 [24.1, 60.0]	2.4 [0.0, 8.3]	0.0
	EMG-only	0.0	0.0	2.0 [0.0, 6.2]

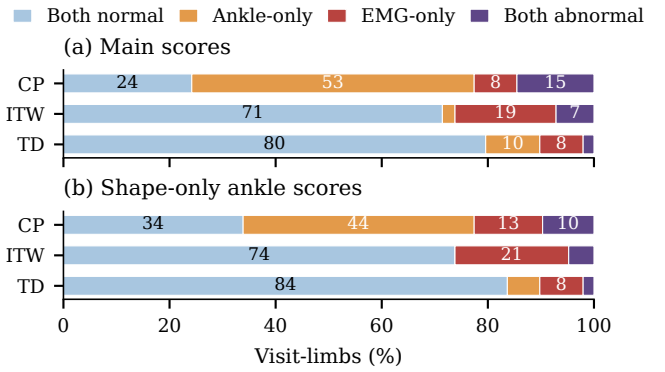


Fig. 2. Four-class concordance per group (standard definition). (a) Main scores; (b) ankle curves mean-centred to remove sensor offset. Numbers are percentages of visit-limbs.

C. Group comparisons and robustness

Table III summarises the tests. The CP ankle-only excess was significant in every analysis and grew under age matching (60.0% versus 10.2%). CP EMG-only abnormality did not differ from TD. The ITW EMG-only excess pointed in the same direction in all analyses but was not significant ($p = 0.11$–0.19). One-vote-per-child estimates were close to the visit-limb rates (CP ankle-only 48.0%, ITW EMG-only 15.9%, TD 10.0% and 8.0%). Within TD, age was not associated with any score ($|\rho| \leq 0.20$, all $p \geq 0.17$). Mean-centring the ankle curves lowered CP ankle-only abnormality to 43.5% (TD 6.1%) under the standard definition and to 12.9% under the

TABLE III
DIFFERENCE FROM TD IN PERCENTAGE POINTS [95% CI] AND PERMUTATION p. AGE-MATCHED: PATIENT VISITS AGED 5.0–12.2 Y (CP 14/19, ITW 9/11 CHILDREN)

Analysis	CP ankle-only	CP EMG-only	ITW EMG-only
Main	+43.0 [19.1, 62.5]	−0.1 [−10.9, 11.3]	+10.9 [−2.1, 23.9]
	$p = 0.001$	$p = 1.00$	$p = 0.18$
Age-matched	+49.8 [23.1, 70.6]	−4.2 [−13.0, 5.0]	+11.3 [−2.6, 25.5]
	$p < 0.001$	$p = 0.45$	$p = 0.19$
Shape-only ankle	+37.4 [14.8, 56.9]	+4.7 [−7.0, 18.1]	+13.3 [−0.5, 26.8]
	$p = 0.003$	$p = 0.51$	$p = 0.13$

robust one, showing that part of the robust kinematic signal reflected sensor offset.

D. Longitudinal pilot

Only seven children had a valid same-cycle pre/post pair (three CP, four ITW; 14 limbs); two used a T2 follow-up and five a T1 follow-up (56–224 days after T0). A further CP child was lost because the affected limb's ankle angle was missing at T0. With the main scores, four limbs in three children were discordant (CP 2/3, ITW 1/4). After mean-centring the ankle curves only one CP child remained discordant, in both limbs (ankle stance score -12.9 and -15.0 degrees). No limb showed an EMG improvement larger than δ_s under either analysis, so the pilot cannot distinguish persistent abnormal activation from changes too small to measure.

V. DISCUSSION

Calibrating against same-day variability changed the picture substantially. Without it, about one visit-limb in ten in TD children and about one in four in CP (22.6%) and ITW (26.2%) would have been labelled EMG-abnormal. With it, none of the EMG-only flags survived. Area-normalised envelope shape, which captures timing but not amplitude, therefore carries limited information beyond its own measurement noise in this wearable two-trial setting. This is consistent with the high within-session variability reported for pediatric EMG [6] and argues for reporting repeatability alongside any EMG-based change score.

The dominant CP pattern, abnormal ankle motion with EMG shape inside the normal range, agrees with reports that kinematic abnormality in CP is not a direct consequence of abnormal activation patterns [5] and that kinematics can improve after botulinum toxin without EMG pattern change [4]. Non-neural factors such as muscle stiffness, contracture and altered joint mechanics are plausible contributors, but our data cannot test this. The ITW trend toward swing-phase gastrocnemius activity matches the premature swing firing previously described in toe walkers [12], but with 11 children it remains a hypothesis.

The sensitivity analysis matters for interpretation. Removing constant ankle offset reduced robust CP ankle-only abnormality from 41.9% to 12.9% and the longitudinal discordance from three children to one. Inertial-sensor placement and calibration differ between sessions, and within-session repeatability does not capture this between-visit error. For treatment tracking, shape-based ankle scores or between-session repeatability data are therefore preferable.

Limitations. The cohorts are small, especially ITW and the seven-child longitudinal pilot, and treatments were heterogeneous. Only two muscles and one joint were available. The repeatability threshold is estimated from single-trial differences while visit scores average two trials, which makes the robust definition conservative. Whole-phase RMSE can dilute brief timing errors, so an onset/offset-based timing measure might be more sensitive. One session is labelled differently in the file name and the participant records, and the detected toe-off varied widely (35–81% of the cycle). A flag indicates measurable discordance; it does not establish treatment failure.

VI. CONCLUSION

A repeatability-calibrated, phase-specific concordance analysis of wearable gait data showed that abnormal ankle kinematics with typical EMG shape characterise about half of CP visit-limbs, a result robust to child-level weighting, age matching and sensor offset. EMG shape deviations, by contrast, rarely exceeded same-day trial-to-trial variability, and the dataset contains few valid same-treatment pre/post pairs. Before EMG shape is used to track pediatric treatment response, studies should collect more trials per session, quantify between-session repeatability and test timing-specific EMG measures. Code for every step is available from the authors on request.

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